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MASTER OF COMPUTER APPLICATION (REVISED)/ BACHELOR OF COMPUTER APPLICATION (REVISED) (MCA/BCA)

Term-End Examination

June, 2024

MCS-013 : DISCRETE MATHEMATICS

Time : 2 Hours *Maximum Marks : 50*

Note : *Question* ***No. 1*** *is compulsory. Attempt any* ***three*** *questions from the rest.*

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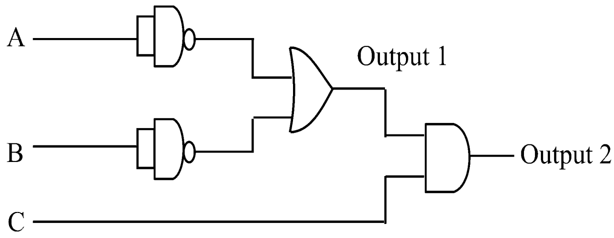
1. (a) Using truth table, show that : 2

$$\sim(p \to q) \equiv p \wedge \sim q$$

(b) Prove that : 2

$$(A - B) \cup B = A \cup B$$

(c) Find the Boolean expression for the outputs of the following circuit : 2



(d) Make Venn diagram for the following set of expressions : 2

(i) $A \cap B \cap C$

(ii) $A \cup B - C$

(e) Find the domain for which the function : 2

$$f(x) = 3x^2 - 1$$

and

$$g(x) = 1 - 5x$$

are not equal.

(f) A coin is tossed n times. What is the probability of getting r heads ? 2

(g) Prove the following : 2

$$\sim (\forall_x \, P(x)) \equiv \exists_x (\sim P(x))$$

(h) If : 2

$$f = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{bmatrix}$$

and

$$g = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{bmatrix},$$

find $f \text{ o } g$ and $g \text{ o } f$.

(i) Use mathematical induction to prove : 2

$$\frac{1}{1\times 2} + \frac{1}{2\times 3} + \ldots\ldots + \frac{1}{n\times(n+1)} = \frac{n}{n+1}.$$

(j) How many different strings can be made from the letters of the word 'SUCCESS', using all the letters ? 2

2. (a) Let R be the relation on the set of strings of Hindi letters such that $a\text{R}b$ iff $l(a) = l(b)$, where $l(x)$ is length of string x. Show that R is an equivalence relation. 5

(b) Write contrapositive, converse and the inverse of the implication "the home team does not win whenever it is raining." 3

(c) Make Pascal's triangle upto $n = 6$. 2

3. (a) Compare conjunctive normal form and disjunctive normal form. Give suitable example for each. 5

(b) Construct the logic circuit for the following Boolean expressions : 5

(i) $(a \wedge b \wedge c) \vee (b \wedge c)' \vee (a \wedge b)'$

(ii) $(a' \wedge b') \vee (b' \wedge c) \vee d$

4. (a) If : 2

$$2\text{P}(n, 2) + 50 = \text{P}(2n, 2)$$

then find n.

(b) Find in how many ways can 25 identical books be placed in 5 identical boxes. 3

(c) Write short notes on any *two* of the following : 5

(i) Modus-Tollens

(ii) Syllogism

(iii) Contrapositive

5. (a) If there are 12 persons in the party, and if each two of them shake hands with each other, how many handshakes happen in the party ? 3

(b) Prove that : 2

$$\mathrm{A} - (\mathrm{A} - \mathrm{B}) \equiv \mathrm{A} \cap \mathrm{B}$$

using Venn diagram.

(c) Reduce the following expression to the simpler form : 2

$$\mathrm{F}(a, b) = (a' \wedge b') \vee (a' \wedge b) \vee (a \wedge b')$$

(d) If : 3

$$P(X) = 0.25,$$

$$P(Y) = 0.4$$

and $P(X \cup Y) = 0.5$,

then determine :

(i) $P(X \cap Y)$

(ii) $P(X \cap Y')$

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MCS-013

M. C. A. (REVISED)/B. C. A. (REVISED) (MCA/BCA)

Term-End Examination

June, 2023

MCS-013 : DISCRETE MATHEMATICS

Time : 2 Hours *Maximum Marks : 50*

***Note :** Question **No. 1** is compulsory. Attempt any **three** questions from the rest.*

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1. (a) Differentiate between predicate logic and propositional logic. Give suitable example to show the difference. 4

   (b) Show that : 4

   $$(p \to \sim q) \wedge (p \to \sim r) \equiv \sim \left[p \wedge (q \vee r)\right]$$

   without using truth table.

   (c) Show whether $\sqrt{5}$ is irrational or rational. 4

   (d) Construct the logic circuit represented by the Boolean expression : 4

   $$\left(x_1' \vee \left(x_2 \wedge x_3'\right)\right) \wedge \left(x_2 \vee x_4'\right)$$

   where x_1, x_2, x_3, x_4 are inputs to the circuit.

   (e) Find $f^{-1}(x)$, where $f(x) = \dfrac{x-2}{x-3}$. 4

2. (a) How many ways are there to rank n candidates for the job of a chef ? In how many rankings will Mr. X be in the second place ? 4

   (b) Prove that : 4

   $$C(n, r) + C(n, r-1) = C(n+1, r)$$

   (c) State Absorption Law and Involution Law of Boolean Algebra. 2

3. (a) Explain the difference between a pair of mutually exclusive events and a pair of independent events. Your explanation should include an example for each. 5

(b) In how many ways can 15 students of MCA and 10 students of BCA be groped into 4 groups ? 3

(c) Let A = {1, 2, 3, 4, 5}. Construct a relation R from A to A such that R is reflexive and symmetric but not transitive. 2

4. (a) A sequence of ten bits (0's and 1's) is randomly generated. What is the probability that at least one of the bits is 0 ? 3

(b) Find the number of permutations of the word ATTENDANT. 3

(c) One red and one white dice are rolled. What is the probability that the red die turns up a smaller number than the white die ? 4

5. (a) Make Venn diagram for each of the situations given below : 4

(i) $A \subseteq B$

(ii) $A \cap B = \phi$

(b) Let $f(x) = \sqrt{x+1}, \quad \forall\, x \geq -1$ and $g(x) = x^3, \ \forall\, x \in R$, then find $f \circ g$ and $g \circ f$. 2

(c) Suppose a valid computer password consists of 8 characters, the first of which is the digit 1, 3 or 5. The rest of the 7 characters are either English alphabets or a digit. How many different passwords are possible ? 4

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MCS-013

M. C. A./B. C. A. REVISED
(MCA/BCA)

Term-End Examination

June, 2020

MCS-013 : DISCRETE MATHEMATICS

Time : 2 Hours *Maximum Marks : 50*

Note : *Question No. 1 is compulsory. Attempt any **three** questions from the rest.*

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1. (a) Check whether the following formula is tautology, contradiction or contingency : 5

$$\sim ((P \rightarrow Q) \rightarrow ((R \vee P) \rightarrow (R \vee Q)))$$

(b) Two finite sets have x and y number of elements. The total number of subsets of the first set is four times the total number of subsets of second set. Find out the value of $x - y$. 4

(c) In a group of 400 people 250 can speak in English only and 70 can speak Hindi only. 3

(i) How many can speak English ?

(ii) How many can speak Hindi ?

(iii) How many can speak both English and Hindi ?

(d) If $f : A \rightarrow B$ and $g : B \rightarrow C$ are injective function, then $g \circ f : A \rightarrow C$ is an injective function. Prove or disprove. 3

(e) Use the method of proof by contradiction to show that $x \in R$ if $x^3 + 4x = 0$, then $x = 0$. 3

(f) Three persons enter in a railway compartment. If there are 5 seats vacant, in how many ways they can take these seats ? 2

2. (a) Given : 5

$$A = \{1, 3, 5, 7\}$$

$$B = \{2, 3, 5, 8\}$$

(i) List the elements of $(A \times B) \times (B - A)$.

(ii) Is $(A \times B) \times (B - A)$ a subset of $A \times B$?

(b) Prove that : 5

$$^{n}C_{r} + {}^{n}C_{r-1} = {}^{n+1}C_{r} \; (0 \le r \le n).$$

3. (a) Show that in any set of eleven integers there are two which are divisible by 10, by applying pigeonhole principle. 3

(b) How many solutions are there of : 4

$$x + y + z = 17$$

subject to the constraints :

$$x \ge 1$$

$$y \ge 2$$

$$z \ge 3.$$

(c) If : 3

$$P(A) = \frac{1}{4}$$

$$P(B) = \frac{2}{5}$$

and $P(A \cup B) = \frac{1}{2}$

find :

(i) $P(A \cap B)$

(ii) $P(A \cap B')$

4. (a) Five balls are drawn from a bag containing 6 white and 4 black balls. What is the probability that 3 are white and 2 black ? 3

(b) From the digit 1, 2, 3, 4, 5, 6, how many three digit odd numbers can be formed when

(i) repetition of digit is allowed ? 2

(ii) repetition of digit is not allowed ? 2

(c) How many numbers divisible by 2 lying between 50,000 and 70,000 can be formed from the digits 3, 4, 5, 6, 7, 8, 9, no digit being repeated in any number. 3

5. (a) Show that : 4

$$1.2 + 2.3 + \dots\dots\dots n(n+1) = \frac{n(n+1)(n+2)}{3}$$

(b) Write the negation of the following statement : 2

If it is raining, then the game is cancelled.

(c) Draw the circuit represented by the following Boolean function : 4

$$f : xy + \bar{x}y$$

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MCS-013

MCA (Revised) / BCA (Revised)

Term-End Examination

June, 2022

MCS-013 : DISCRETE MATHEMATICS

Time : 2 hours *Maximum Marks : 50*

Note : *Question no. **1** is **compulsory**. Attempt any **three** questions from the rest.*

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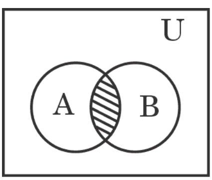
1. (a) Use pigeonhole principle to find the minimum number of cars that will have same colour; if 20 cars are painted using 6 colours. *2*

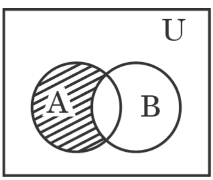
(b) Show that $\sqrt{3}$ is irrational. *4*

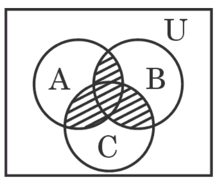
(c) Draw the logic circuit for the following Boolean expression : *4*

$$(X_1 \wedge X_2 \vee X_3') \vee (X_1' \vee X_2' \wedge X_3)$$

(d) Write the set expression (shaded part) for the following Venn diagrams : *3*







(e) Write suitable mathematical statement that can be represented by $(\forall x)(\forall y)(\exists z)P$. *2*

(f) Disprove that
$(\forall a \in R)(\forall b \in R)[(a^2 = b^2) \Rightarrow (a = b)]$,
using a counter-example. *2*

(g) There are 5 women and 6 men. A committee has to be formed of 3 people. In how many ways can this be done if at least one woman should be there in the committee ? *3*

2. (a) Prove using mathematical induction that

$$(1 + x)^n > 1 + nx \text{ for all } x > 0 \text{ and } n > 1. \qquad 5$$

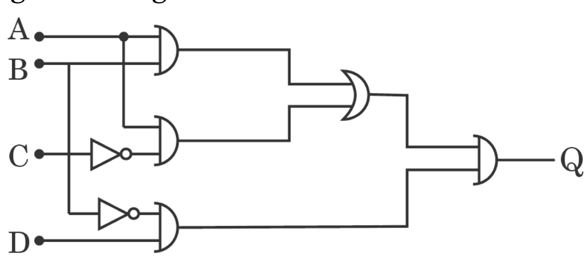
(b) What is proper subset ? Explain with the help of an example. *2*

(c) Show $p \wedge (p \vee q) \leftrightarrow p$ is a tautology. *3*

3. (a) Every function is a relation. Is every relation a function ? Why ? *3*

(b) Explain circular permutation with the help of an example. *3*

(c) Find the Boolean expression of Q in the logic circuit given below : *4*



4. (a) If A and B are two mutually exclusive events such that $P(A) = 0 \cdot 4$ and $P(B) = 0 \cdot 2$, what is the probability that either A or B does not occur ? *2*

(b) Give the geometric representation of $\{3\} \times R$. *2*

(c) Present a direct proof of the statement "Square of an odd integer is odd." *3*

(d) Find the inverse of the function

$$f(x) = \frac{x - 4}{x - 3}.$$ *3*

5. (a) Show that

$$C(n, m)\, C(m, k) = C(n, k)\, C(n - k, m - k)$$

$$\text{for } 1 \leq k \leq m \leq n.$$ *3*

(b) How many 3-letter words can be formed from the letters of the word "HAPPY" ? *2*

(c) What is contradiction ? Show whether $p \wedge q \wedge \sim p$ is a contradiction or not. *3*

(d) Draw the Venn diagram for the expression

$A \cup B \cap C$. *2*

MCA (Revised) / BCA (Revised)

Term-End Examination

June, 2021

MCS-013 : DISCRETE MATHEMATICS

Time : 2 hours *Maximum Marks : 50*

Note : *Question no. **1** is **compulsory**. Attempt any **three** questions from the rest.*

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1. (a) Write the truth value of the conjunction of : "The earth is round" and "3 > 4". *2*

(b) Use Mathematical Induction to prove that :

$$1 + \frac{1}{4} + \frac{1}{9} + \ldots \frac{1}{n^2} \leq 2 - \frac{1}{n} \ \forall \ n \in N.$$ *4*

(c) If $f : R \rightarrow R$ is a function such that $f(x) = 3x - 2$, prove that f is injective. Also find the inverse of f. *5*

(d) Show that $p \vee (q \wedge r)$ and $(p \vee q) \wedge (p \vee r)$ are logically equivalent. *3*

(e) A and B are two mutually exclusive events such that P(A) = 0·4 and P(B) = 0·2. What is the probability that :

(i) A does not occur ?

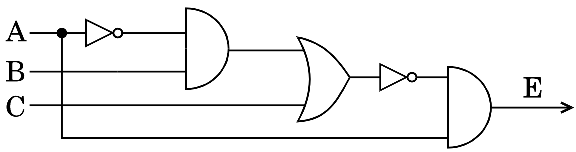
(ii) A or B does not occur ?

(iii) Either A or B does not occur ? *3*

(f) Find the number of ways of placing n people in n – 1 rooms, no room being empty. *3*

2. (a) What is integer partition ? Write down all the partitions of 8. Also find P_8^4 and P_8^7. *4*

(b) Find Boolean Expression for the following logical circuit : *4*



(c) Let two functions be such that $f(x) = x^2 + 2$ and $g(x) = 2x$. Define fog and gof. *2*

3. (a) Reduce the following Boolean Expression to simpler form : *4*

$$E(X_1, X_2, X_3) = (X_1 \wedge X_2 \wedge X_3) \vee (X_1 \wedge X_2) \vee (X_2 \wedge X_3)$$

(b) Show that $\sim(p \to q) \to p$ is a tautology. *2*

(c) Prove that $\sqrt{2}$ is irrational. *4*

4. (a) What is Relation ? How is relation different from function ? Explain any two properties of relations with an example. *5*

(b) A company has the following professionals : Project Leaders – 5, Team Leaders – 6, System Architects – 3.

Find how many different committees can be formed of 10 professionals, each containing at least 2 Project Leaders, at least 3 Team Leaders and at least 1 System Architect. *3*

(c) Find the dual of $A \cup B \cap C$. *2*

5. (a) Explain the Identity Laws of Boolean Algebra. *2*

(b) State and prove the Addition Theorem of Probability. *4*

(c) Verify that $p \wedge q \wedge \sim p$ is a contradiction. *2*

(d) What is Exclusive Disjunction ? Write truth table for $p \oplus q$. *2*